

Deep Learning for Network Packet Forensics Using TensorFlow

TensorFlow is an open source Python library for machine learning. It does mathematical computation using dataflow graphs. This article dwells on the use of TensorFlow as a forensic tool for classifying and predicting malware sourced from honeypots and honeynets.

Data mining and machine learning are key methods used to gain information from databases in which hidden patterns are analysed from a huge repository of records. In classical methodology, there are a number of algorithms for clustering, association rule mining, visualisation and classification to get meaningful information for predictive analysis.

Nowadays, a number of tools and technologies are available for the implementation of data mining and machine learning including WEKA, Tanagra, Orange, ELKI, KNIME, RapidMiner, R and many others which have Java, Python or C++ for back-end programming and customisation.

Machine learning has core tasks associated with classification and recognition, which are usually related to artificial intelligence. Generally, these operations are performed using some metaheuristic approach in which global optimisation or simply effective results can be fetched from a huge search space of solutions.

There are a number of prominent soft computing approaches such as neural networks, fuzzy logic, support vector machines, swarm intelligence, metaheuristics, etc. Metaheuristics, in turn, has many optimisations like Ant Colony Optimisation, Cuckoo Search, Bees Algorithm, Particle Swarm Optimisation, etc.

Despite the number of metaheuristic approaches and other effective soft computing algorithms, there are many applications for which a higher degree of accuracy and lower error rate is required.

For machine learning, the approaches of artificial neural networks (ANN) or support vector machines (SVM) can be used with the dataset to be integrated. ANN can be used for malware detection or classification, face recognition, fingerprint or finger vein structure analysis in which the previous dataset is used for training a model and then the prediction or classification of further datasets is done. ANN based learning is fully dependent on the dataset used for training the model and apparently, if the dataset is not accurate, the predictive analysis will affect the accuracy of the results.

To implement modelling and training with ANN, there are a number of open source tools available including SciLab, OpenNN, FANN, PyBrain and many others.

Deep learning

Deep learning is one of the branches of machine learning with a strong base of algorithms that have multi-layered processing, a higher degree of computations and accuracy, and a lower error rate with the integration of deep graph based learning. Other branches of deep learning include deep structured learning, deep machine learning or hierarchical learning.

